



CSE 200 • Group Presentation

VARIATIONAL AUTOENCODER

Learning to generate, not just compress

Presented by **Member 1** • **Member 2** • **Member 3**

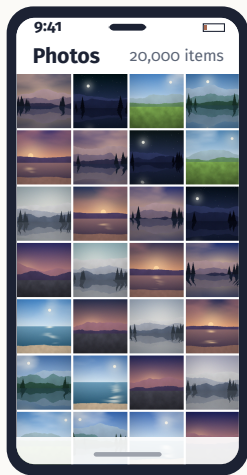
Department of CSE, BUET



latent space $z \sim \mathcal{N}(\mu, \sigma^2)$



The Problem: Storage Almost Full



20,000 photos

Imagine your phone has **20,000 photos**, but the storage is **almost full**.

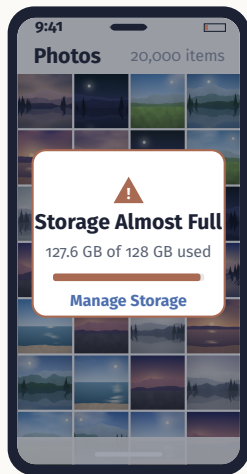
Storage

127.6 GB of 128 GB used



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB

The Problem: Storage Almost Full



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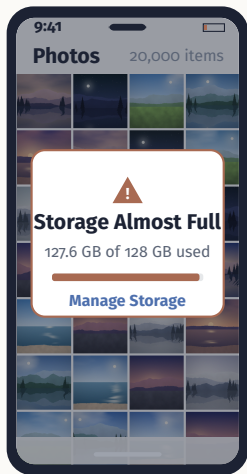
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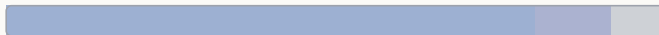


20,000 photos

Imagine your phone has **20,000 photos**, but the storage is **almost full**.

Storage

127.6 GB of 128 GB used



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB **only 0.4 GB free**

Delete memories?



not ideal: once gone,
they're gone forever

A smarter solution

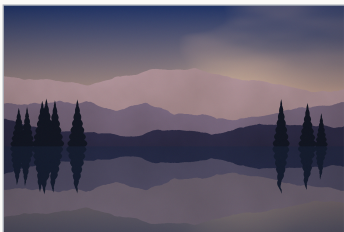


keep every photo, but
use far less space

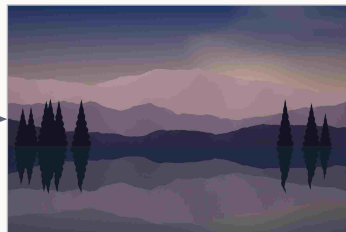
A Smarter Solution: Compress



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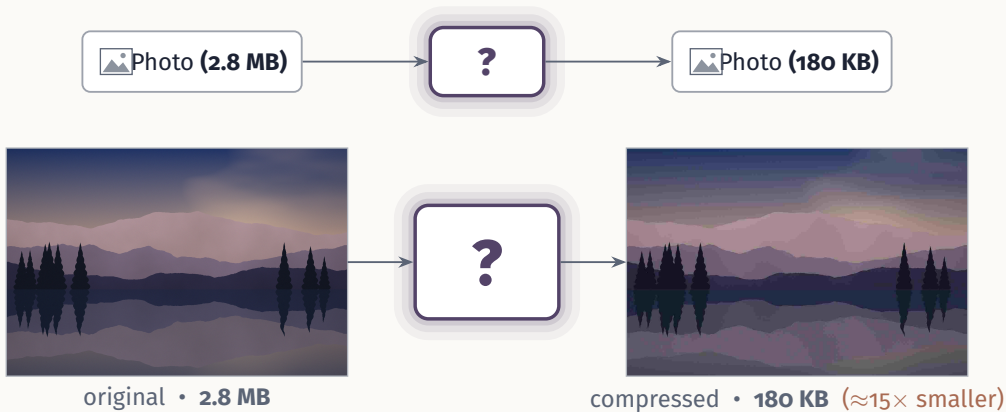


original • 2.8 MB



compressed • 180 KB ($\approx 15\times$ smaller)

A Smarter Solution: Compress

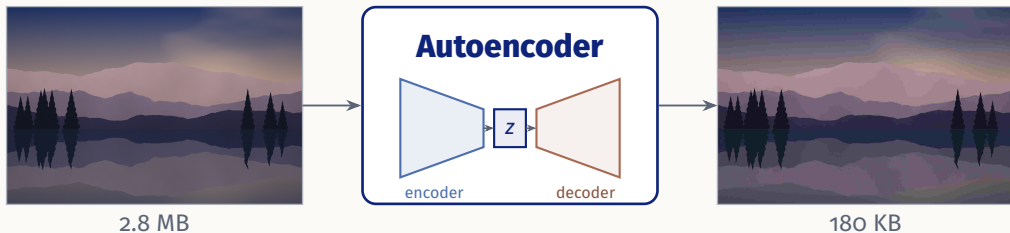


Same photo, much less storage. What goes inside the **mysterious middle box**?

Enter the Autoencoder

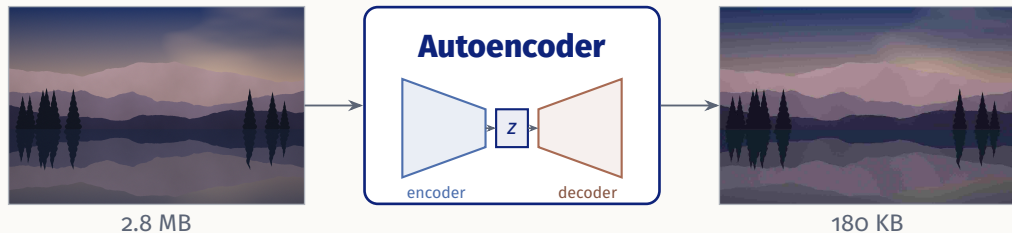


Enter the Autoencoder



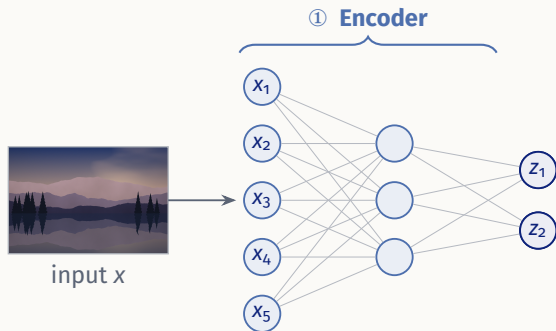
This box is exactly where **autoencoders** come in.

Enter the Autoencoder



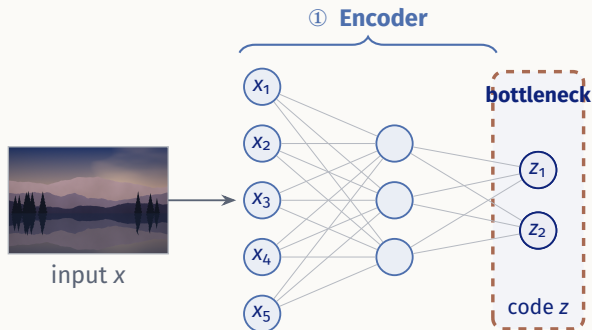
This box is exactly where **autoencoders** come in.
a neural network that learns to **compress** an image and **rebuild** it

Autoencoder Architecture



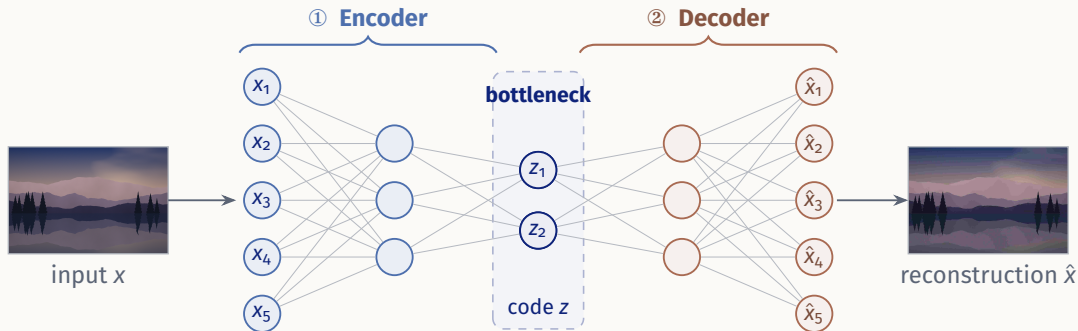
The **encoder** compresses the image, layer by layer, until it reaches the bottleneck.

Autoencoder Architecture



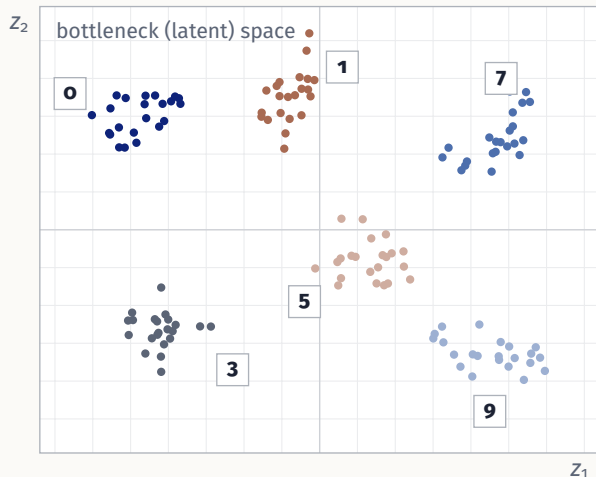
The **bottleneck** is a narrow doorway: only the *essential, non-redundant* information can pass.

Autoencoder Architecture



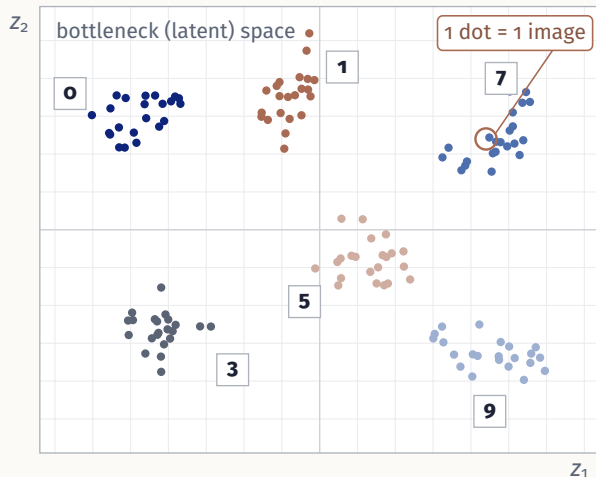
The **decoder** uses that compact code to rebuild the photo: $\hat{x} \approx x$.

The Problem with Autoencoders



Each point is **one image** after the autoencoder has compressed it to the bottleneck.

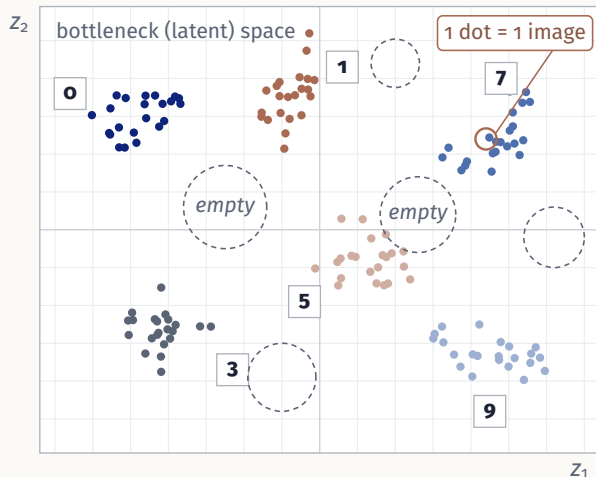
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Similar images land close together: the points form separate **islands**.

The Problem with Autoencoders

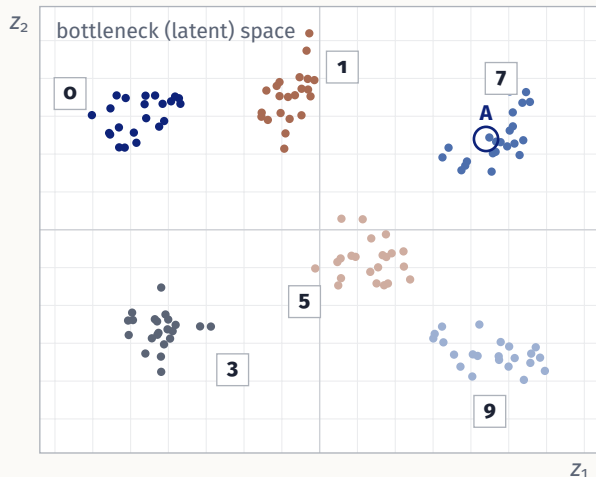


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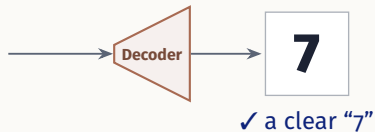
Similar images land close together: the points form separate **islands**.

Between the islands there are **large empty gaps**: no training image was ever placed there.

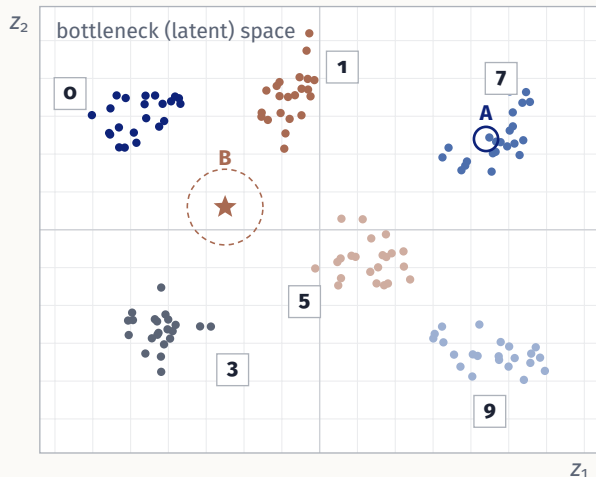
What Lives in the Gaps?



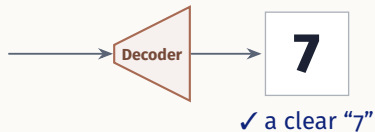
A point inside an island



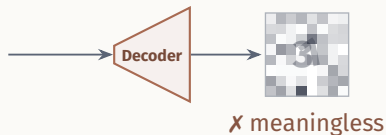
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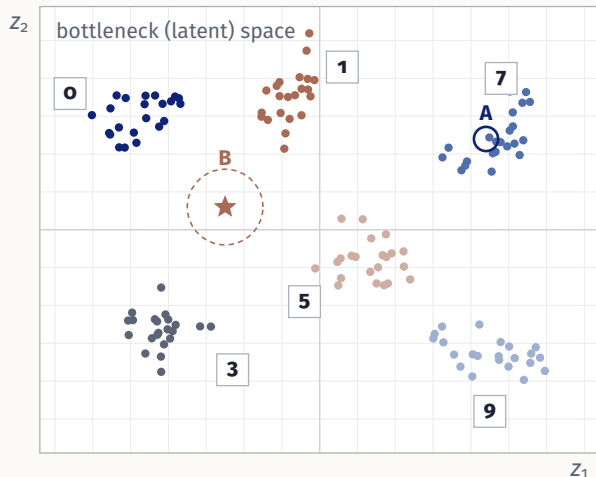
A point inside an island



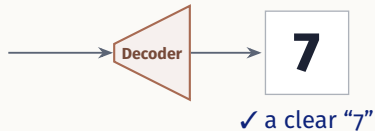
B random point from an empty gap



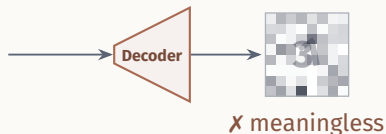
What Lives in the Gaps?



A point inside an island

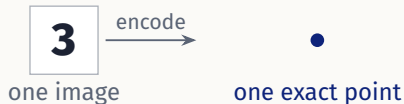
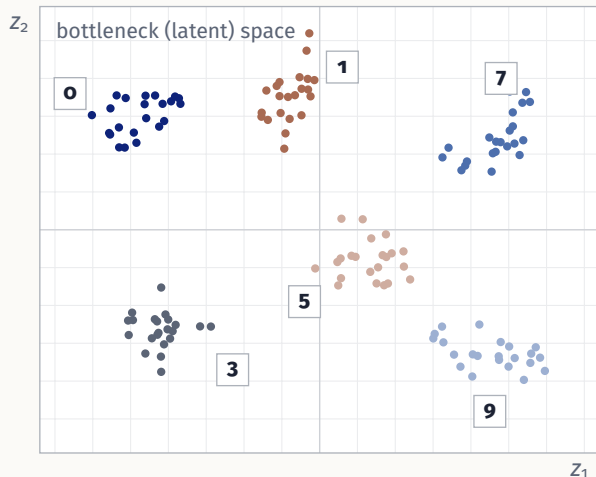


B random point from an empty gap



What if every image were stored as a **distribution** instead of one exact point?
⇒ the core idea behind the **Variational Autoencoder**

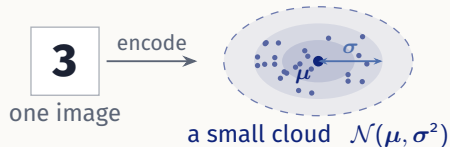
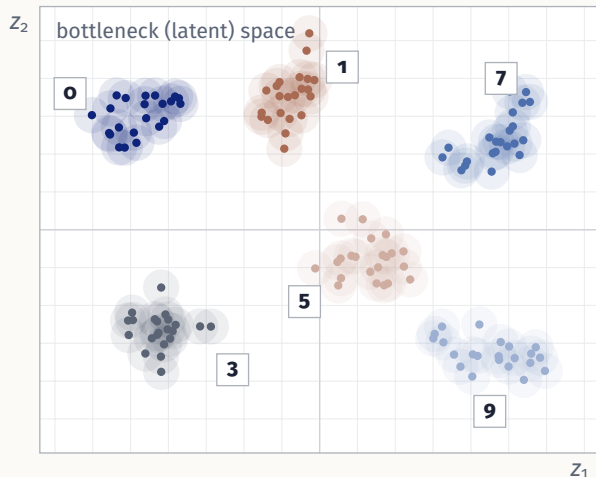
From Points to Distributions



Autoencoder: one image $\rightarrow$ one point

Right now every image is a single point, with empty gaps in between.

From Points to Distributions

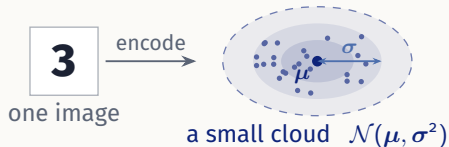
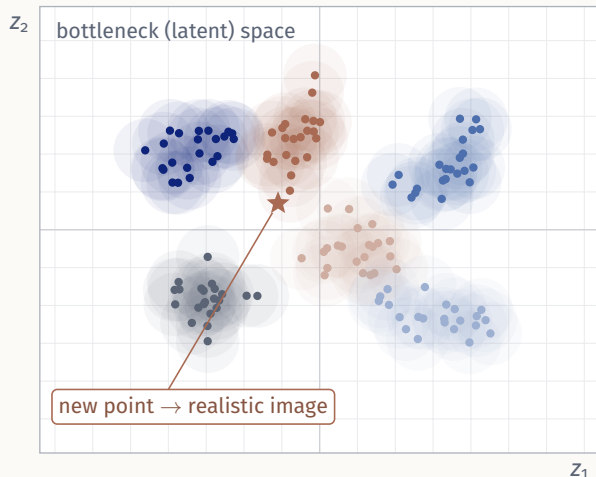


Autoencoder: one image $\rightarrow$ one point

VAE: one image $\rightarrow$ one **distribution**

Each image now occupies a **small area** instead of a single point.

From Points to Distributions

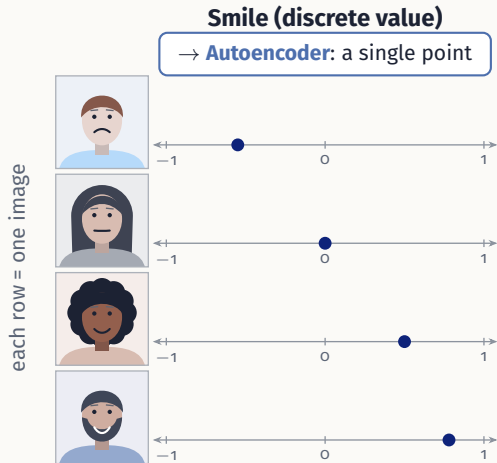


Autoencoder: one image $\rightarrow$ one point

VAE: one image $\rightarrow$ one **distribution**

The areas overlap, so **the gaps disappear**: we can pick a new point from that area and still generate a realistic new image.

One Image, One Distribution

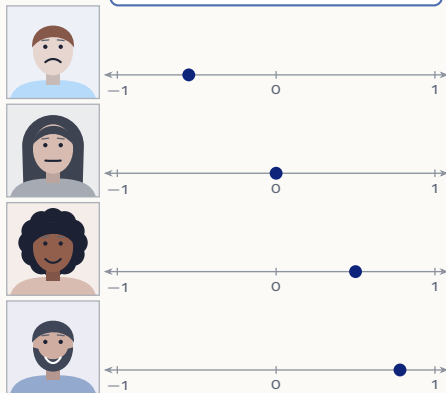


One Image, One Distribution

each row = one image

Smile (discrete value)

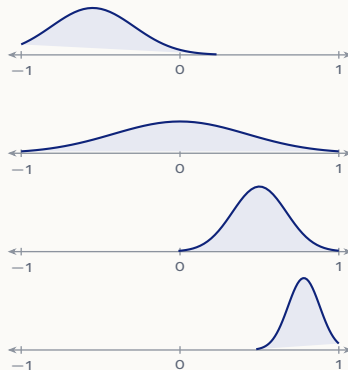
→ **Autoencoder**: a single point



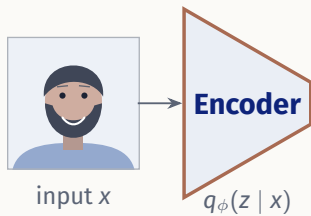
Smile (probability distribution)

→ **Variational Autoencoder**: a distribution

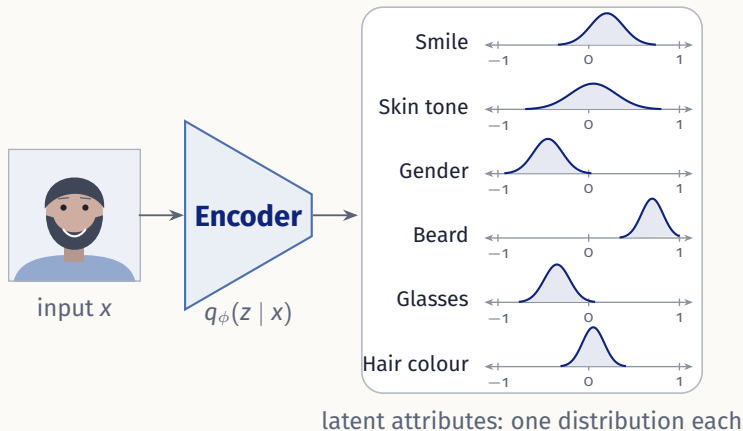
vs.



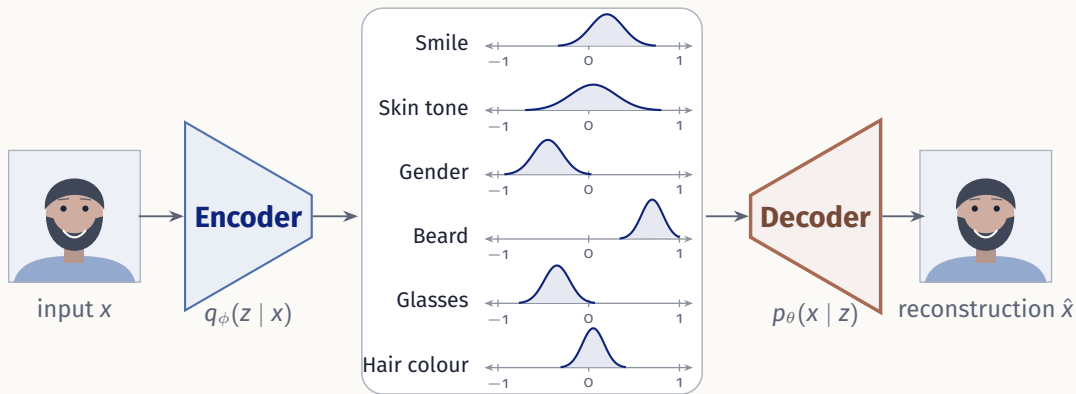
VAE Simplified: Latent Attributes



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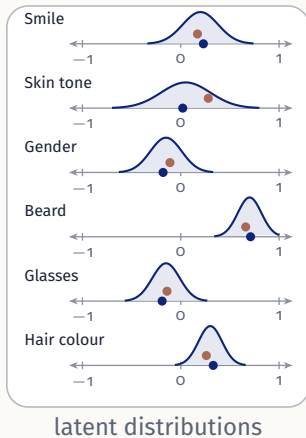


VAE Simplified: Latent Attributes

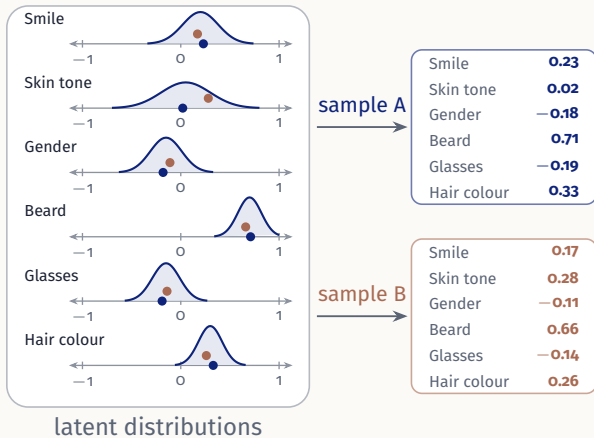


latent attributes: one distribution each

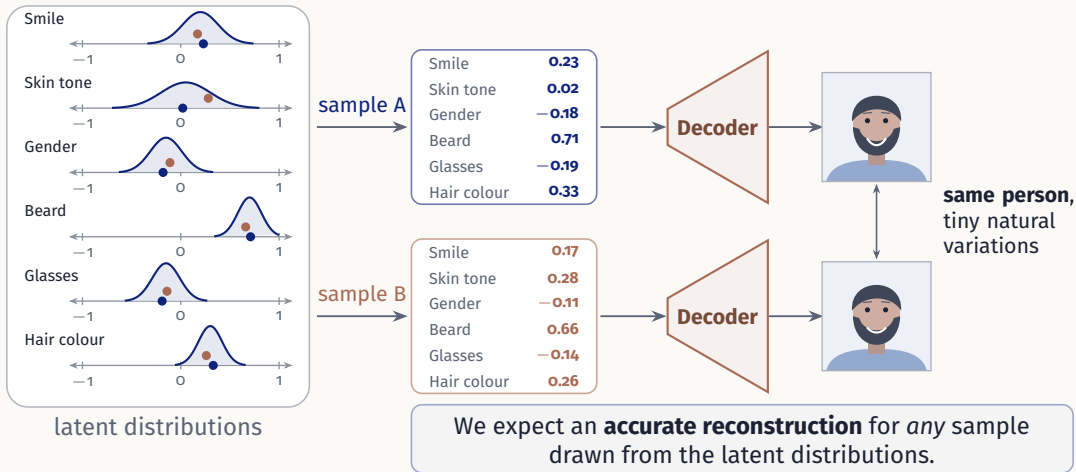
VAE Simplified: Sample, then Decode



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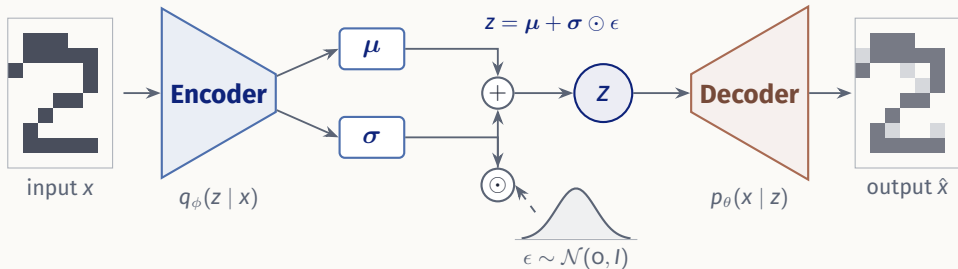


VAE Simplified: Sample, then Decode



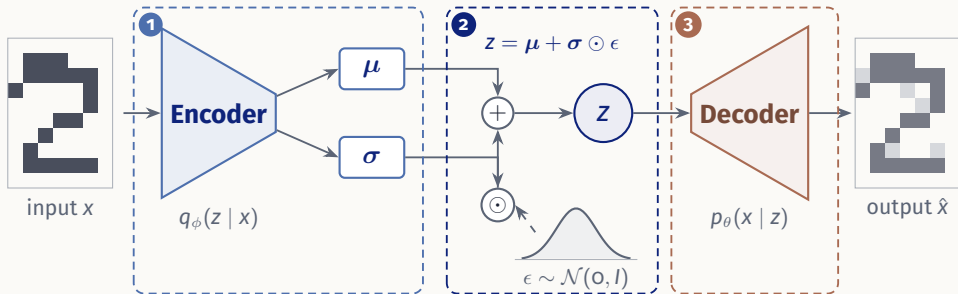
Up Next: The VAE Workflow

So far: **Autoencoder** = one point per image • **VAE** = one distribution per image



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Handing over to **Member 2**: ① encoder $\rightarrow$ ② sampling $z \rightarrow$ ③ decoder